From Endogenous Credit to Borrower Composition
A Reproducible Measure from Japanese Sectoral Loan Stocks

Abstract

Theories of endogenous credit distinguish credit creation from allocation, while public statistics often report aggregates or borrower-sector stocks. We construct a reproducible borrower-composition measure from Bank of Japan (BOJ) sectoral loan stocks. The primary mapping implements Bezemer et al. (2020)'s published Japan crosswalk: non-financial business, financial business, property/mortgage, and household non-housing credit. Scale and composition use the same credit population. Stock changes are aggregated within each taxonomy bucket before taking positive parts, making the four-quarter vector invariant to finer classifications that preserve those buckets. Construction belongs to non-financial business in the primary crosswalk and moves in two literature-anchored alternatives based on Werner (1997) and Müller & Verner (2024). Evidence centers on the BOJ mapping, exact identities, coverage and residual accounting, positive and net changes, and joint alternative allocations. Ministry of Finance statistics provide borrower-side convergence for manufacturing; Ministry of Land, Infrastructure, Transport and Tourism purpose data partially validate household housing. A current-vintage pseudo-out-of-sample exercise is only an application: all main-table augmentations have higher point loss than the matched-stock baseline. One Appendix row improves point loss, but its interval crosses zero. The contribution is a verifiable measurement bridge, not a purpose measure or forecasting result.

Keywords: endogenous credit; borrower composition; sectoral lending; measurement; Bank of Japan

JEL classification: E44; E51; E58; G21; G28

1 Introduction

Theories of endogenous money and the monetary circuit explain why bank lending cannot be reduced to a prior stock of loanable funds (Le Bourva, 1992; Fontana, 2003; McLeay et al., 2014). The use and institutional location of credit also matter: a loan to a non-financial firm, a construction company, a financial intermediary, or a household enters a different balance-sheet position even when the aggregate amount is unchanged (Rochon & Rossi, 2007; Passarella, 2014; Werner, 2012). Empirical work nevertheless often measures credit scale without an equally explicit measure of its composition.

Existing studies show that mortgage and business credit (Bezemer et al., 2020; Bezemer & Zhang, 2019), tradable and non-tradable sector credit (Müller & Verner, 2024; Andries et al., 2025), and the riskiness of corporate credit allocation (Brandão-Marques et al., 2022) have different macro-financial implications. Earlier work for Japan also documents shifts and dispersion in sectoral loan growth (Saita & Sekine, 2001). These contributions motivate disaggregation but do not solve the measurement problem considered here. The Bank of Japan (BOJ) reports detailed outstanding loans by borrower sector, not comprehensive gross originations jointly classified by borrower and purpose. A stock change can reflect new lending, repayment, write-off, or reclassification. A borrower label does not identify what the loan financed.

This paper therefore makes a narrower claim than a destination-of-credit paper. It constructs and audits an observable borrower-composition measure. Its primary specification follows the published Japan crosswalk of Bezemer et al. (2020), which divides the included population into non-financial business (NFB), financial business, property/mortgage, and household non-housing credit. This replaces an author-defined three-way classification as the main measure. The scalar q_t is only the NFB coordinate of the four-part vector. It is a borrower-composition statistic, not a share of credit reaching a final destination.

The classification literature does not imply a unique placement for construction. It is part of NFB in the primary Japan crosswalk, enters the financial-circulation proxy in Werner (1997), and is non-tradable in Müller & Verner (2024). We implement all three placements over the same included credit population and report them jointly. This design converts an otherwise hidden author choice into a disclosed robustness dimension. The current-sample declaration is not an externally time-stamped preregistration; the earlier author-defined G/B/E grouping is retained only in the Appendix for archive compatibility.

The paper makes three contributions. First, it uses the early endogenous-credit literature to motivate two observables—credit scale and borrower composition—without claiming that borrower labels reveal loan use. Second, it makes the published crosswalk, included population, residuals, coverage, aggregation order, positive-versus-net convention, and construction placements fully auditable. Third, it provides a falsifiable application protocol: external MOF and MLIT data partially validate selected branches, while a pseudo-out-of-sample (OOS) comparison shows how the bridge can be used. These checks do not validate the full four-bucket construct or establish incremental predictability.

2 Conceptual Motivation and Measurement Design

2.1 From endogenous credit to two observable coordinates

Classical quantity theories relate nominal expenditure to money and its velocity (Fisher, 1911; Friedman, 1956; Laidler, 1985). Endogenous-credit accounts instead start from bank lending and deposit creation (Le Bourva, 1992; Fontana, 2003; McLeay et al., 2014). The Quantity Theory of Credit further emphasises that the economic use of credit conditions its transmission (Werner, 2012). Monetary-circuit theory supplies an institutional account: bank credit opens and finances circuits whose balance-sheet positions need not be equivalent (Rochon & Rossi, 2007; Passarella, 2014).

The available BOJ data do not identify those uses. They do identify borrower sectors and two household-purpose components. Let B_t^{TOT} be the BOJ total for domestically licensed banks, where "domestically licensed" describes the lenders rather than borrower geography. Let B_t^{FIN}, B_t^{RE}, B_t^{LG}, B_t^{HH}, and B_t^{HOU} denote, respectively, finance and insurance, real estate, local-government, total household, and household housing stocks. The primary Japan crosswalk defines

$$L_t^{NFB} = B_t^{TOT} - B_t^{FIN} - B_t^{RE} - B_t^{LG} - B_t^{HH}, \tag{1}$$

$$L_t^{FIN} = B_t^{FIN}, \tag{2}$$

$$L_t^{PROP} = B_t^{RE} + B_t^{HOU}, \tag{3}$$

$$L_t^{HHN} = B_t^{HH} - B_t^{HOU}. \tag{4}$$

The included stock is therefore

$$L_t^P = \sum_{k \in \mathcal{K}} L_t^k = B_t^{TOT} - B_t^{LG}, \qquad \mathcal{K} = \{NFB, FIN, PROP, HHN\}. \tag{5}$$

The published NFB residual contains the separately disclosed overseas-linked series and a small unresolved remainder. We retain both explicitly in the coverage audit rather than relabelling them as domestic borrower observations. The scale coordinate used at horizon h is

$$g_{t,h}^{P} = \Delta_h \log L_t^P. \tag{6}$$

Thus scale and composition refer to the same lenders, loan instrument, and borrower population.

For composition, first form each bucket's net change and positive part,

$$\Delta L_t^k = L_t^k - L_{t-1}^k, \qquad P_t^k = \max\{\Delta L_t^k, 0\}, \qquad C_t = \sum_{\ell \in \mathcal{K}} P_t^\ell. \tag{7}$$

For a window of w quarters, define

$$P_{t,w}^k = \sum_{j=0}^{w-1} P_{t-j}^k, \qquad \pi_{t,w}^k = \frac{P_{t,w}^k}{\sum_{\ell \in \mathcal{K}} P_{t,w}^\ell}. \tag{8}$$

The primary empirical object is the four-quarter vector $\boldsymbol{\pi}_{t,4} = (\pi_{t,4}^{NFB}, \pi_{t,4}^{FIN}, \pi_{t,4}^{PROP}, \pi_{t,4}^{HHN})$. The scalar $q_t \equiv \pi_{t,4}^{NFB}$ is a display coordinate. Four-quarter aggregation reduces quarter-specific stock-difference noise and remains defined whenever the four-quarter positive-flow denominator is positive. Neither the vector nor q_t is a share of credit used for production, investment, or asset purchases.

2.2 Granularity invariance and claim boundary

Suppose a component of L_t^k is divided into additive subseries $s_1, \ldots, s_m$ that remain in bucket k. Then

$$\sum_{j=1}^{m} \Delta L_{s_j,t} = \Delta L_{s,t}.$$

Because the positive part in equation (7) is applied after this sum, P_t^k, $\boldsymbol{\pi}_{t,w}$, and q_t are unchanged. The measure is therefore invariant to classification refinements within a fixed borrower bucket. It is not invariant to moving a sector across buckets. That remaining classification uncertainty is addressed by three literature-anchored mappings reported jointly rather than by presenting a single author choice as uniquely implied by the data.

This construction differs from $\sum_{s \in \mathcal{S}_k} \max\{\Delta L_{s,t}, 0\}$. The latter rises when offsetting expansions and contractions are published as separate subseries and therefore depends on classification granularity. The empirical audit reports the difference between the two orders rather than silently treating them as equivalent.

The measurement design imposes five checks before any outcome application is considered. First, the mapped stocks should approximately reconcile with the published BOJ aggregate after the local-government exclusion is made explicit. Second, the four buckets and the scale stock must exactly exhaust the same included population. Third, within-bucket refinements should leave the primary measure unchanged. Fourth, positive and net changes should be reported separately because they answer different accounting questions. Fifth, cross- bucket ambiguity should be evaluated under mappings fixed from the literature. A purpose interpretation would additionally require an independent source that codes loan purpose. The current same-provider comparator cannot supply that validation.

Table 1: Concept-to-measure mapping and claim-boundary checks.

Object	Observable	Required check	Supported interpretation
Credit scale	$g_{t,h}^{P} = \Delta_h \log L_t^P$	exact same-population stock identity	growth of included BOJ loan stocks
Borrower composition	$\boldsymbol{\pi}_{t,4}$, with $q_t = \pi_{t,4}^{NFB}$	four buckets, sum-then-positive	composition of positive bucket-level stock changes
Net reallocation	$(\Delta L_t^k)_{k\in\mathcal{K}}$	retain expansions and contractions	signed changes, not bounded shares
Taxonomy robustness	three literature-anchored placements	same population; jointly reported	sensitivity to cross-bucket assignment
Household residual	BOJ consumer-purpose stock	same-provider level comparison	limited check, not external validation
Outcome use	expanding-window loss comparison	matched-scale baseline and uncertainty interval	application, not construct validation

3 Data and Empirical Design

3.1 BOJ source, population, and timing

The source is the BOJ Time-Series Data Search, database LA01. The selected map is the total outstanding loans of domestically licensed banks less local-government loans. The lender designation does not restrict borrowers to Japan: the published NFB residual includes the disclosed overseas-linked series. BOJ documentation states that the sector data have a discontinuity in 2009Q2 following an industrial-classification revision (Bank of Japan, n.d.). A change ending in 2009Q2 crosses that break, so the first valid within-taxonomy change is 2009Q3. The first complete four-quarter measure is dated 2010Q2 and becomes usable in the application in 2010Q3 under an assumed fixed 90-day release lag.

The long-term JGB yield, nominal GDP, and BOJ total assets are proxy outcomes used only to demonstrate the bridge; none measures borrower composition or validates it. Table 5 reports the JGB application, nominal GDP is not tabulated, and BOJ balance-sheet acceleration is confined to the Appendix.

3.2 Mapping and coverage

Table 2 is the empirical centre of the paper. It reports the published borrower labels, construction of the four primary buckets, explicit exclusion, embedded residual components, and each bucket's share of positive changes. The primary population is $L_t^P = B_t^{TOT} - B_t^{LG}$, not a domestic-borrower subset. Accounting coverage reconciles that population and the explicit local-government exclusion to the official aggregate. Decomposition coverage separately shows how much of published residual NFB is accounted for by detailed NFB industries, the overseas-linked series, and the unresolved remainder. Neither coverage concept establishes loan purpose.

Table 3 makes construction placement central rather than incidental. The primary Bezemer–Samarina–Zhang crosswalk places construction within NFB. Werner's Japan operationalisation places construction with real estate and non-bank financial institutions in a financial-circulation category; our BOJ adaptation uses the available broader finance-and-insurance series and is labelled a proxy. The Müller–Verner sector crosswalk places construction among non-tradables; its implementation here is likewise a BOJ-sector adaptation. The three measures exhaust the same included population; only their categorical partition changes. They are therefore informative about classification sensitivity, not alternative estimates of unobserved loan purpose.

Table 2: Literature-anchored BOJ credit-allocation crosswalk.

BOJ input / component	Bridge role	Construction	Main measurement risk	Primary / explicit flow share
Published Japan residual NFB	P_t^{NFB}	Official total less finance/insurance, real estate, local government, and total household loans; construction is included.	The residual also absorbs the separately disclosed overseas-linked and unresolved components.	40.1%
Finance and insurance	P_t^{FIN}	Direct BOJ borrower-sector match to financial business.	Borrower sector does not identify the use of funds.	17.9%
Real estate plus household housing	P_t^{PROP}	Published Japan property/mortgage crosswalk.	Combines a borrower-sector stock with a household-purpose stock.	40.0%
Household total less household housing	P_t^{HHN}	Published residual construction for household non-housing credit.	Includes tax and other household borrowing; it is not labelled consumer credit.	2.1%
Local governments	excluded, explicit	Removed by the published Japan residual formula.	Exclusion changes the population relative to the official total.	6.0%
Overseas-linked loans	NFB component, explicit	Disclosed because the current BOJ total makes it part of residual NFB.	Not a domestic borrower-sector observation.	5.9%
Unresolved residual	NFB component, explicit	Residual after mapped NFB industries and overseas-linked loans.	Has no independent borrower or purpose label.	0.0%

The primary four-bucket mapping follows Bezemer, Samarina, and Zhang (2020; Japan crosswalk in DNB Working Paper 559); `q_t` is the four-quarter NFB coordinate. Construction is an NFB detail. Overseas-linked and unresolved amounts absorbed by the published residual are shown explicitly. For the four primary buckets, the displayed share is $\sum_t P_t^k / \sum_t (P_t^{NFB} + P_t^{FIN} + P_t^{PROP} + P_t^{HHN})$ over the 67 valid within-taxonomy quarterly changes (2009Q3–2026Q1). Explicit component rows use the denominator stated by their construction. Positive parts are taken after net changes are aggregated within each fixed bucket. The current-sample mappings are literature-anchored and reported jointly; this declaration is not an externally time-stamped preregistration.

Table 3: Literature-anchored credit-taxonomy placements.

Taxonomy identifier	Role	Population	Construction placement	Selection rule
bezemer_samarina_zhang_2020_japan_v1	primary	Total outstanding loans of domestically licensed banks less local-government loans, exactly exhausted by non-financial business, financial business, property/mortgage, and household non-housing. Domestically licensed describes the lenders, not borrower geography; the residual NFB series includes the explicit overseas-linked component.	Construction is an NFB detail in the primary crosswalk; it is not a stand-alone purpose bucket.	literature-anchored mappings reported jointly; the current-sample declaration is not an externally time-stamped pre-registration
werner_1997_financial_circulation_v1	literature_anchored_robustness	Exactly the included population of bezemer_samarina_zhang_2020_japan_v1.	Construction enters the Werner-inspired financial-circulation proxy; the available BOJ finance/insurance series is broader than Werner's original non-bank-financial-institution component.	literature-anchored mappings reported jointly; the current-sample declaration is not an externally time-stamped pre-registration
muller_verner_2024_sector_v1	literature_anchored_robustness	Exactly the included population of bezemer_samarina_zhang_2020_japan_v1.	Construction is nontradable, following the published tradable/nontradable sector crosswalk.	literature-anchored mappings reported jointly; the current-sample declaration is not an externally time-stamped pre-registration
author_gbe_legacy_v1	appendix_legacy	Mapped legacy G/B/E borrower groups.	Construction is separate in this legacy grouping.	Appendix compatibility only.

The primary and robustness mappings are defined in replication metadata and reported jointly. The current-sample declaration is not an externally time-stamped preregistration; the prospective archive freezes these definitions only for future releases.

3.3 Pre-outcome measurement audits

Aggregation order and granularity. For each taxonomy, the primary calculation aggregates signed changes within every fixed bucket before applying the positive part and then forms a four-quarter flow-weighted vector. Splitting or combining additive series within a bucket therefore cannot change the result. Moving construction or another sector across buckets can change it; Tables 2 and 3 disclose those placements rather than hiding them in an author-defined label.

Positive and signed changes. The bounded shares in equation (8) describe the composition of positive bucket-level net changes. The signed vector $(\Delta L_t^k)_{k\in\mathcal{K}}$ retains contractions and reconciles to the change in the same included stock. A signed contribution ratio can fall outside $[0,1]$ when bucket changes have opposing signs, so it is not reported as a composition share. Keeping both objects prevents repayment or contraction from being silently reinterpreted as positive allocation.

Purpose and borrower-side comparators. The BOJ consumer-purpose household stock is compared with $L_t^{HHN} = B_t^{HH} - B_t^{HOU}$. It does not define the residual and is not an input to q_t. Because it comes from the same provider and covers a narrower concept than household non-housing credit, it is a limited consistency check, not independent external purpose validation. Source-independent checks are also reported: MOF corporate statistics provide debtor-side convergent evidence for the manufacturing branch of NFB, and MLIT purpose-classified originations provide direct-purpose evidence for the household-housing branch of PROP. Neither source validates the full four-bucket composition.

Alternative allocations. The Werner-inspired and Müller–Verner mappings provide literature-anchored alternatives to the primary crosswalk, with construction placed differently in each. All are implemented on the same credit population and reported jointly. The current archive records these definitions, but it does not turn them into an externally time-stamped preregistration. The resulting coordinates are classification-robustness checks, not data-driven weights, confidence bounds, or attempts to recover unobserved purpose.

4 Empirical Application

4.1 Same-population reconciliation and coverage

The primary four buckets exactly exhaust the included stock, and their signed changes exactly exhaust the change in that stock. The maximum recorded gap is zero at the precision reported in Table 4. Werner and Müller–Verner coordinates also exhaust this same population. This resolves a scale-composition population mismatch, but it is an accounting result rather than evidence about loan use.

The included stock has a median 94.5% coverage ratio relative to the official total; the explicit local-government exclusion accounts for a median 5.5%, and the scope identity has zero maximum gap. Within published residual NFB, detailed NFB industries account for 94.3% at the median, the disclosed overseas-linked component for 5.7%, and the unresolved component for 0.001%. Their decomposition identity also has zero maximum gap. These readouts keep coverage, geography, and the unexplained residual visible instead of treating published NFB as a purely domestic industry aggregate.

4.1.1 Positive and net changes

The positive-flow vector and the net-change vector are retained as distinct objects. Taking the positive part after within-bucket aggregation prevents finer publication detail from mechanically increasing a bucket's measured flow. The four positive coordinates share one denominator and sum to one to the precision reported. Net changes preserve contractions and exactly reconcile to the included stock change; they are not forced into bounded shares. The primary four-quarter vector is available for 64 of 67 valid within-taxonomy changes, beginning in 2010Q2. The three unavailable endpoints reflect the rolling construction and source availability, not an inferred borrower state.

4.1.2 Literature-anchored mappings and limited validation

Replacing a private classification with a published primary crosswalk and literature-anchored robustness mappings reduces discretion but does not remove taxonomy uncertainty. Construction is the clearest test: the three literature-based placements differ even though the underlying BOJ population is fixed. The appropriate robustness claim is therefore that results can be compared under these disclosed partitions, not that one taxonomy identifies the true economic destination.

The household non-housing residual is non-negative in every reported row and its stock identity has zero maximum gap. Its correlation with the narrower BOJ consumer-purpose stock is 0.904 over 67 paired observations, with a median absolute stock gap of 35,947 in units of JPY 100 million. This is consistent with a related household-credit component, but level correlation from the same provider is not source-independent purpose validation. The stronger destination claim therefore remains unsupported. Table 6 adds partial source-independent evidence without upgrading that claim.

Table 4: Published-taxonomy BOJ bridge audits.

Audit	Current readout	Interpretation
Primary same-population identity	max |C_t-sum four buckets|=0.000000; max |included stock-sum four stocks|=0.000000; max |signed-scale gap|=0.000000	Allocation and scale are constructed from one included population. This is an accounting identity, not purpose validation.
Published residual NFB decomposition	median mapped-NFB/NFB=94.3%; median overseas/NFB=5.7%; median unresolved/NFB=0.001%; max decomposition gap=0.000000	The primary NFB series follows the published Japan residual exactly. Overseas-linked and unresolved components absorbed by that residual are visible rather than silently treated as mapped domestic industries.
Household non-housing split	negative residual rows=0; max split-identity gap=0.000000; consumer-purpose pair N=67, corr=0.904, median absolute stock gap=35947 (JPY 100m)	Household non-housing is total household credit less housing credit. The narrower consumer-purpose series is a same-provider comparator, not independent external validation.
Explicit official-population coverage	median included/official stock=94.5%; median local-government/official=5.5%; max explicit-scope gap=0.000000	The primary population plus the explicit local-government exclusion reconciles to the official total; coverage does not validate loan purpose.
Literature-anchored taxonomy population identity	Werner max population gap=0.000000; Muller-Verner max population gap=0.000000	The Werner-inspired and Muller-Verner BOJ adaptations are defined in metadata and exhaust the same included population. They are reported jointly; this is not a claim of externally time-stamped preregistration.
Primary four-quarter vector availability	valid q_t=64/67; first=2010-06-30; max |vector sum-1|=0.000000; max |q_t-NFB alias gap|=0.000000	q_t is only the displayed NFB coordinate of the four-part, common-denominator borrower-composition vector.

4.2 Use example: pseudo-OOS loss comparison

For horizon $h \in \{4, 8\}$, the use example augments the matched BOJ credit- stock-growth baseline with a reported taxonomy coordinate $z_t^{(r)}$:

$$y_{t,t+h} = \alpha_t + \beta_t g_{t,h}^P + \gamma_t z_t^{(r)} + \varepsilon_{t+h}, \tag{9}$$

where r denotes the Bezemer–Samarina–Zhang NFB coordinate, the Werner financial-circulation coordinate, or the Müller–Verner non-tradable coordinate. Thus Table 5 uses the bridge without treating the three classifications as interchangeable.

The design uses current-vintage data with assumed fixed release lags and does not reconstruct historical vintages. Labels whose forward outcomes are not realised by a forecast origin are purged. Predictors are standardised within each expanding training window. The moving-block bootstrap uses a block length equal to the horizon and 2,000 fixed-seed replications. Its interval is for the mean difference in standardised squared-error loss, not for the difference in RMSE.

The taxonomy, candidate feature, baseline, horizons, and minimum training size are fixed before each outcome comparison. At every forecast origin, fitting and standardisation use the training window only. A specification changed after its holdout loss is inspected is a new model version and requires later data; the same holdout cannot be reused as validation.

Table 5 is an illustration of how a borrower-composition coordinate can be added to a scale-only benchmark. It is not construct validation, a causal design, or a forecasting contribution. All six main-table loss differentials are positive, so every displayed composition-augmented model has worse point loss than the matched-stock baseline. At four quarters, the primary and Werner intervals include zero, whereas the Müller–Verner interval is above zero; at eight quarters, all three intervals are above zero. The short post-break sample, overlapping horizons, current-vintage inputs, and conditional bootstrap intervals preclude a claim of incremental predictability. The single negative point differential appears only in the Appendix and its interval includes zero.

4.3 Partial external validation

The MOF comparison uses four-quarter changes in manufacturing corporations' short- and long-term bank borrowing and BOJ manufacturing loan stocks over 2010Q2–2026Q1. Pearson correlation is 0.735, Spearman correlation is 0.792, and change directions agree in 70.3% of 64 common observations. These are borrower-side convergence statistics across non-identical populations; they do not validate purpose or the full NFB bucket. The MLIT FY2024 survey directly classifies individual housing-loan originations: new and existing acquisition account for 93.5%, refinancing 6.4%, and apartment-loan originations total JPY 3.82 trillion. This validates the availability of a housing-purpose margin for the household-housing branch of PROP, not corporate real-estate purpose or quarterly BOJ stock changes (Ministry of Finance, 2026; Ministry of Land, Infrastructure, Transport and Tourism, 2026).

4.4 Implications for data design

A destination or purpose measure would require gross originations classified jointly by borrower and use, separately reporting repayments, write-offs, refinancing, and reclassifications. It would also require stable taxonomies, revision vintages, and reconciliation from detailed cells to a published aggregate. Source-independent validation would additionally require purpose-coded data matched as closely as possible to the same lender, instrument, borrower, and timing population. Until such data are assembled, a borrower-composition release should ship with its literature source, code map, common start, aggregation order, positive and net versions, residual decomposition, and jointly

Table 5: Bridge application: JP borrower-composition pseudo-OOS loss comparison for the long-term JGB yield change.

h	Literature-anchored coordinate	N	Coordinate-aug. RMSE	Matched-stock RMSE	Mean Δ squared loss [95% CI]
4Q	Bezemer: NFB (primary)	24	1.787	1.735	0.185 [-0.494, 0.912]
4Q	Werner-inspired BOJ proxy	24	1.859	1.735	0.446 [-0.110, 1.079]
4Q	Müller–Verner: non-tradable (BOJ adaptation)	24	1.973	1.735	0.881 [0.354, 1.475]
8Q	Bezemer: NFB (primary)	12	3.121	2.735	2.260 [1.021, 3.517]
8Q	Werner-inspired BOJ proxy	12	3.362	2.735	3.821 [0.662, 6.980]
8Q	Müller–Verner: non-tradable (BOJ adaptation)	12	3.468	2.735	4.545 [3.215, 5.890]

Rows use the current-vintage Japan pseudo-OOS panel with an assumed fixed 90-day lag for BOJ inputs. Scale is log growth of the exact Bezemer Japan-crosswalk population (BOJ total less local-government loans) in primary__included__stock; the lender population is domestically licensed banks, but the residual NFB bucket includes the disclosed overseas-linked series. Each displayed four-quarter coordinate uses the same population. The three literature-anchored mappings are reported jointly rather than choosing among their displayed loss results: Bezemer non-financial business, a Werner-inspired BOJ borrower-sector proxy, or a BOJ-sector adaptation of Muller-Verner non-tradables. Positive parts are taken after aggregation within each reported taxonomy bucket. The 2009Q2 cross-taxonomy flow is invalid; dated BOJ inputs are shifted by that assumed fixed lag. Targets report standardized RMSE. Each h-quarter regression excludes training labels whose outcomes are not realized by the forecast origin and requires at least 28 complete training cases in both models. The moving-block length equals the horizon, with 2,000 fixed-seed replications. The mean loss differential is candidate standardized squared loss minus matched-stock baseline loss; it is not an RMSE difference. Its block-bootstrap interval is descriptive and conditional on the displayed construction and training-window choice, not confirmatory post-selection inference. Negative values favor the candidate. Reported forecast origins range from 2018-06-30 to 2024-03-31.

The Bezemer non-financial-business share, four-quarter coordinate is available in 64/67 post-break source quarters (missing-source pattern Q1:1,Q3:1,Q4:1). The Werner-inspired BOJ borrower-sector proxy share, four-quarter coordinate is available in 64/67 post-break source quarters (missing-source pattern Q1:1,Q3:1,Q4:1). The Muller-Verner non-tradable share, four-quarter coordinate is available in 64/67 post-break source quarters (missing-source pattern Q1:1,Q3:1,Q4:1).

Only models augmenting matched-stock growth are shown; the companion CSV contains standalone predictors, identity checks, simple benchmarks, and alternative borrower-composition constructions when available. No p-values or multiplicity-adjusted inference are reported.

For the primary Bezemer four-quarter row, the 4Q mean loss differential is 0.131 with a 20-case minimum and 0.145 with a 24-case minimum; all three signs are above zero, so none of these window choices shows a point-loss improvement.

Each row augments matched-stock growth with the displayed coordinate. The table is a use case for the BOJ borrower-composition bridge, not a validated forecasting model.

Table 6: Partial external validation of published-bucket branches.

External source	Primary branch	Estimand	Period and readout	Claim boundary
MOF corporate survey	NFB: manufacturing	Correlation of debtor-side and BOJ lender-side 4Q changes in manufacturing borrowing stocks.	2010Q2-2026Q1; N=64; Pearson=0.74; Spearman=0.79; direction=70.3%	Borrower-side convergent evidence only; populations and lender coverage differ, and no loan purpose is observed.
MLIT private housing-loan survey	PROP: household housing	Published purpose composition of individual housing-loan originations and apartment-loan amounts.	FY2024; acquisition=93.5%; refinancing=6.4%; apartment originations=JPY 3.82tn	Direct purpose only for the annual housing branch; it does not validate corporate real-estate loans or quarterly BOJ stock changes.

MOF uses four-quarter differences to reduce sensitivity to its annual April–June sample replacement; this does not make the two credit populations identical. MLIT is annual origination evidence and is therefore not correlated with the quarterly stock-change bridge.

reported literature-anchored alternatives. Those items form the transferable measurement protocol of this paper.

5 Limits

The measure has clear boundaries:

- It observes the sector of the borrower, not the purpose, destination, recipient, collateral, or final expenditure financed by a loan.
- Positive stock changes are not gross originations. Repayment, refinancing, write-offs, and reclassification can change the stocks.
- Granularity invariance holds for refinements within a fixed bucket, not for moving a sector between reported mappings; construction makes this dependence visible.
- Published residual NFB includes an overseas-linked component and a small unresolved component; both are disclosed but not independently classified.
- The consumer-purpose comparator is narrower than household non-housing credit and is not source-independent; MOF and MLIT evidence validates only selected branches under non-identical populations.
- Assumed fixed release lags do not remove look-ahead from historical data revisions.
- The application sample is too short for confirmatory predictive or causal inference.

6 Conclusions

This paper constructs a four-quarter measure of borrower composition from Japanese sectoral loan stocks. The primary measure follows a published four-bucket Japan crosswalk, while two literature-anchored robustness mappings expose how construction placement changes the coordinate without changing the credit population. The central evidence is the BOJ mapping, exact same-population identities, explicit coverage and residual accounting, positive and net changes, the limited household comparison, and jointly reported alternative allocations. These checks support a reproducible borrower-composition bridge, not a destination or loan-purpose interpretation. Table 5 demonstrates one use of the bridge against a matched-stock baseline. All main-table point losses are worse. One Appendix row improves point loss, but its interval crosses zero. The current sample therefore does not establish incremental predictability.

Declarations

Data availability. The empirical results use public macro-financial data sources described above. An anonymised replication archive accompanies the double-blind review files. After acceptance, the authors intend to release the code, bridge tables, and replication scripts in a public repository.

Conflicts of interest. The authors declare no conflicts of interest.

Declaration of generative AI use. OpenAI Codex was used to assist with language editing and manuscript organisation. The authors reviewed and edited all resulting content and take full responsibility for the content of the article.

Appendix A: Reproducibility and Data Sources

Reproducibility. The anonymised review archive regenerates the BOJ mapping, validation audits, and pseudo-OOS outputs from the supplied processed inputs. It hashes the inputs and compares regenerated files byte-for-byte with expected snapshots. The archive does not refetch the BOJ source data.

Detailed mapping. The anonymous replication archive supplies the code-level map, series labels, release timing, common-availability information, and all alternative assignments. Table 7 records the detailed published-taxonomy crosswalk and its explicit residuals.

Table 7: Published-taxonomy BOJ crosswalk and explicit residuals.

BOJ input / source	Bridge role	Construction	Remaining measurement risk	Flow share
Published Japan residual NFB	P_t^{NFB}	Official total less finance/insurance, real estate, local government, and total household loans; construction is included.	The residual also absorbs the separately disclosed overseas-linked and unresolved components.	40.1%
Finance and insurance	P_t^{FIN}	Direct BOJ borrower-sector match to financial business.	Borrower sector does not identify the use of funds.	17.9%
Real estate plus household housing	P_t^{PROP}	Published Japan property/mortgage crosswalk.	Combines a borrower-sector stock with a household-purpose stock.	40.0%
Household total less household housing	P_t^{HHN}	Published residual construction for household non-housing credit.	Includes tax and other household borrowing; it is not labelled consumer credit.	2.1%
Local governments	excluded, explicit	Removed by the published Japan residual formula.	Exclusion changes the population relative to the official total.	6.0%
Overseas-linked loans	NFB component, explicit	Disclosed because the current BOJ total makes it part of residual NFB.	Not a domestic borrower-sector observation.	5.9%
Unresolved residual	NFB component, explicit	Residual after mapped NFB industries and overseas-linked loans.	Has no independent borrower or purpose label.	0.0%
BOJ consumer-purpose loans	validation comparator	Checks, but does not define, the household non-housing residual.	Same provider and narrower concept; not independent external validation.	validation audit
Bezemer, Samarina, and Zhang (2020), Journal of Banking & Finance 113, 105760; Japan crosswalk in DNB Working Paper 559.	literature source	Taxonomy definitions are literature-anchored and all three are reported jointly.	A published crosswalk does not eliminate BOJ measurement error.	not selected on fit

The four primary flow coordinates share one included population. Local government is outside that population. Overseas-linked and unresolved amounts are disclosed as components absorbed by the published NFB residual, not silently discarded.

Appendix B: Legacy Author-Defined Grouping

The earlier G/B/E specification is retained only for reproducibility and archive compatibility. It is not the primary classification and is not used to select the literature-anchored mappings reported in the main text.

Table 8: Detailed BOJ source series to borrower-composition buckets.

BOJ series name / code	Borrower sector	Borrower bucket	Grouping rationale	Ambiguous cases	Type	Expected bias direction	Negative quarterly changes	Assumed release lag	Series share (incl. U)
Outstanding loans by sector, 19 author-defined Group G series: 19 codes in replication metadata; examples: DLLILKG21 / DLLI5DS2TMK; DLLILKG86 / DLLI5DS2TAF; DLLILKG61 / DLLI5DS2TRO	Manufacturing, trade, services, utilities, transport, local governments, and other organisations	Group G: broad sectors, local governments, and other organisations	Author-defined broad borrower grouping used to construct the composition share.	Borrower sector does not reveal whether funds finance operations, fixed assets, refinancing, or financial assets.	sectoral stock-difference	Measures borrower composition, not loan purpose or macroeconomic use.	Sector stocks are summed within the bucket before the positive part; the signed audit keeps the bucket net change.	Quarterly BOJ LA01 release; forecast panels apply release-lag dating.	34.4%
Construction outstanding: DLLILKG26 / DLLI5DS2TCT	Construction	Group B: construction	Author-defined construction borrower group kept separate without fractional allocation.	Building finance, land purchase, working capital, and refinancing are not separated.	sectoral stock-difference	A separate borrower bucket does not identify construction loan purpose.	The bucket net change is computed before the positive part; the signed audit keeps the sign.	Quarterly BOJ LA01 release; forecast panels apply release-lag dating.	5.6%
Finance, real-estate, and household outstanding sectors: DLLILKG49 / DLLI5DS2TFI; DLLILKG50 / DLLI5DS2TFX; DLLILKG62 / DLLI5DS2TPN	Finance and insurance; real estate; households	Group E: finance, real estate, and households	Author-defined grouping of the remaining mapped domestic borrower sectors without a common purpose assignment.	Real-estate, finance, housing, and household-consumption uses are mixed.	sectoral stock-difference	The group is heterogeneous and must not be read as an asset-purpose measure.	Sector stocks are summed within the bucket before the positive part; the signed audit keeps the bucket net change.	Quarterly BOJ LA01 release; forecast panels apply release-lag dating.	54.2%
Unclassified outstanding sector: DLLILKG63 / DLLI5DS2TFL	Overseas yen loans and domestic loans transferred overseas	Excluded from domestic borrower composition	The reported category is not a domestic borrower-sector group comparable with G, B, or E.	Could contain many borrower sectors and loan uses outside the domestic map.	sectoral stock-difference	Exclusion narrows the mapped domestic denominator; direction for the Group G share is unknown.	Its bucket-positive change enters the selected-series coverage denominator; the signed audit keeps the sign.	Quarterly BOJ LA01 release; forecast panels apply release-lag dating.	5.8%
New loans for fixed investment by sector: 24 codes in replication metadata; examples: DLLILKG21 / DLLI5DS5TMK; DLLILKG86 / DLLI5DS5TAF; DLLILKG63 / DLLI5DS5TFL	Same sector map as the outstanding-loan series	Within-BOJ purpose-coded comparator	Provides a purpose-coded fixed-investment comparator within the same BOJ sector map.	Excludes working capital, refinancing, rollovers, and many asset purchases.	claim-boundary audit	Narrower than total lending and tilted toward investment-intensive borrowers.	Used as reported and kept separate from the borrower-stock-change measure.	Quarterly BOJ LA01 release; forecast panels apply release-lag dating.	audit only

Shares use positive parts taken after net changes are aggregated within borrower buckets. Stock levels are jointly observed from 2009Q2; the first valid within-taxonomy change is 2009Q3. They are selected-series composition shares, not coverage of the official BOJ aggregate.

Appendix C: Asset-Boom Hypothesis and BOJ Balance-Sheet Acceleration

Exploratory asset-boom hypothesis. A separate literature links broad credit and leverage expansions to asset-price cycles and financial instability (Borio & Lowe, 2002; Jordà et al., 2015; Schularick & Taylor, 2012). The borrower groups used here do not identify asset-directed lending. The balance-sheet outcome below therefore records an earlier application question without testing or supporting an asset-boom hypothesis.

The auxiliary specification applies the same matched-scale design to forward changes in the BOJ balance sheet. It is retained to disclose the previously examined use case, not as evidence that the borrower groups identify asset- directed lending. The only negative point-loss differential is the Müller–Verner four-quarter row, -0.097, with a 95% interval of $[-0.223, 0.010]$; the interval crosses zero. Every other row has a positive point-loss differential.

Table 9: Auxiliary BOJ balance-sheet acceleration pseudo-OOS check.

h	Literature-anchored coordinate	N	Coordinate-aug. RMSE	Matched-stock RMSE	Mean Δ squared loss [95% CI]
4Q	Bezemer: NFB (primary)	24	1.338	0.886	1.004 [-0.032, 2.178]
4Q	Werner-inspired BOJ proxy	24	1.356	0.886	1.054 [-0.266, 2.983]
4Q	Müller–Verner: non-tradable (BOJ adaptation)	24	0.974	0.886	0.164 [-0.135, 0.543]
8Q	Bezemer: NFB (primary)	12	0.492	0.299	0.153 [-0.011, 0.329]
8Q	Werner-inspired BOJ proxy	12	0.812	0.299	0.570 [0.339, 0.802]
8Q	Müller–Verner: non-tradable (BOJ adaptation)	12	0.266	0.299	-0.019 [-0.045, 0.009]

Rows use the current-vintage Japan pseudo-OOS panel with an assumed fixed 90-day lag for BOJ inputs. Scale is log growth of the exact Bezemer Japan-crosswalk population (BOJ total less local-government loans) in primary_included_stock; the lender population is domestically licensed banks, but the residual NFB bucket includes the disclosed overseas-linked series. Each displayed four-quarter coordinate uses the same population. The three literature-anchored mappings are reported jointly rather than choosing among their displayed loss results: Bezemer non-financial business, a Werner-inspired BOJ borrower-sector proxy, or a BOJ-sector adaptation of Muller-Verner non-tradables. Positive parts are taken after aggregation within each reported taxonomy bucket. The 2009Q2 cross-taxonomy flow is invalid; dated BOJ inputs are shifted by that assumed fixed lag. Targets report standardized RMSE. Each h-quarter regression excludes training labels whose outcomes are not realized by the forecast origin and requires at least 28 complete training cases in both models. The moving-block length equals the horizon, with 2,000 fixed-seed replications. The mean loss differential is candidate standardized squared loss minus matched-stock baseline loss; it is not an RMSE difference. Its block-bootstrap interval is descriptive and conditional on the displayed construction and training-window choice, not confirmatory post-selection inference. Negative values favor the candidate. Reported forecast origins range from 2018-06-30 to 2024-03-31.

The Bezemer non-financial-business share, four-quarter coordinate is available in 64/67 post-break source quarters (missing-source pattern Q1:1,Q3:1,Q4:1). The Werner-inspired BOJ borrower-sector proxy share, four-quarter coordinate is available in 64/67 post-break source quarters (missing-source pattern Q1:1,Q3:1,Q4:1). The Muller-Verner non-tradable share, four-quarter coordinate is available in 64/67 post-break source quarters (missing-source pattern Q1:1,Q3:1,Q4:1).

Only models augmenting matched-stock growth are shown; the companion CSV contains standalone predictors, identity checks, simple benchmarks, and alternative borrower-composition constructions when available. No p-values or multiplicity-adjusted inference are reported.

Each row augments matched-stock growth with the displayed coordinate. This appendix-only specification records the asset-acceleration check separately from the main borrower-composition application.

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